Power Budget - Arm Motion Control (Rev C)

Team Number: 107
Project Name: Simple Educational Robotic Arm - Arm Motion Control subsystem
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Version: Rev C (2026-10-30), matches schematic Rev C

Blue = value from a datasheet or design choice (source in column I). Black = formula. Yellow = result to check.

A. All major components (active devices and ICs; power sources, regulators and passives are listed below)

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit	Source / note
	Dual H-bridge motor driver (channel A used)	onsemi FAN8100N	VCC 2.2-9 V, PVCC 1.8-9 V	1	1500.0	1500.0	mA	FAN8100N datasheet abs. max: peak output current 1.5 A per channel; VCC/PVCC abs. max 10.5 V; Pd 2.0 W.
	DC gearmotor 250:1 Micro Metal MP 6 V	Pololu #2370	6 V nominal	1	670.0	670.0	mA	Pololu #2370: stall current 0.67 A at 6 V (extrapolated); no-load 70 mA.
	PIC18F57Q43 Curiosity Nano (MCU board)	Microchip DM164150	USB 5 V in; target 1.8-5.5 V (3.3 V set)	1	500.0	500.0	mA	Curiosity Nano HW user guide: MIC5353 target regulator max 500 mA; VBUS PTC fuse 500 mA hold.
	LED 3 mm green (power, DS1)	Kingbright WP710A10SGD	Vf 2.2 V typ	1	25.0	25.0	mA	Kingbright datasheet: DC forward current 25 mA max. Runs at about 2.8 mA through R1 = 1 kOhm.
	LED 3 mm yellow (debug, DS2)	Kingbright WP710A10YD	Vf 2.1 V typ	1	20.0	20.0	mA	Kingbright datasheet: DC forward current 20 mA max. Runs at about 3.6 mA through R10 = 330 Ohm.

Rail tables below use the worst-case current each part can draw in THIS circuit (motor stall, LEDs set by their resistors), not the part's absolute rating.

+6 V Motor Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit	Source / note
	DC gearmotor (stall)	Pololu #2370	6 V	1	670.0	670.0	mA	Worst case = stall, 0.67 A.
	FAN8100N output stage (PVCC)	onsemi FAN8100N	PVCC 1.8-9 V	1	65.0	65.0	mA	Datasheet ICC4 total supply current in brake, max 65 mA (taken all on PVCC, conservative).
	LM317T program divider R11 + R12	240 Ohm / 910 Ohm 1 %	6.0 V	1	5.2	5.2	mA	1.25 V / 240 Ohm = 5.2 mA; also serves as LM317 minimum load.
			Subtotal			**740.2**	mA	
			Safety Margin			25%		
			Total Current Required on +6V Rail			**925.3**	mA	

Regulator	+6 V adjustable regulator (VR1)	ST LM317T	Vin 3-40 V above Vout; Vout 1.25-37 V	1	1500.0	1500.0	mA	ST LM317 datasheet: 1.5 A output current. Needs heatsink (Aavid 577102B00000G).
	Total Remaining Current Available on +6V Rail					**574.8**	mA	

+5 V Logic Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit	Source / note
	FAN8100N logic supply (VCC)	onsemi FAN8100N	VCC 2.2-9 V	1	8.0	8.0	mA	Datasheet ICC2 VCC current in brake, max 8 mA.
	Power LED DS1 + R1	Kingbright WP710A10SGD	5 V	1	2.8	2.8	mA	(5 V - 2.2 V) / 1 kOhm = 2.8 mA.
			Subtotal			**10.8**	mA	
			Safety Margin			25%		
			Total Current Required on +5V Rail			**13.5**	mA	

Regulator	+5 V regulator (VR2)	ST L7805CV	Vin 7-35 V	1	1500.0	1500.0	mA	ST L78 datasheet: 1.5 A output current.
	Total Remaining Current Available on +5V Rail					**1486.5**	mA	

+3.3 V Rail (Curiosity Nano, from USB)	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit	Source / note
	PIC18F57Q43 core + peripherals	PIC18F57Q43	1.8-5.5 V	1	25.0	25.0	mA	Design allowance for the MCU at 64 MHz with ADC/PWM running (estimate, well above typical).
	Debug LED DS2 + R10 (RE0)	Kingbright WP710A10YD	3.3 V	1	3.6	3.6	mA	(3.3 V - 2.1 V) / 330 Ohm = 3.6 mA.
	Ribbon outputs GRIP_CMD, ARM_IN_POS	330 Ohm series + 10 kOhm pull-down on far board	3.3 V	2	0.3	0.6	mA	3.3 V / (330 + 10 k) Ohm = 0.32 mA each.
	Expansion header J4 3.3 V pin	J4 (6-pin)	3.3 V	1	50.0	50.0	mA	Allowance reserved for an add-on on J4 (design choice).
			Subtotal			**79.2**	mA	
			Safety Margin			25%		
			Total Current Required on +3.3V Rail			**99.0**	mA	
Regulator	+3.3 V LDO on the Curiosity Nano	Microchip MIC5353	Vin 2.6-6 V	1	500.0	500.0	mA	Curiosity Nano HW user guide: MIC5353 max 500 mA load.
	Total Remaining Current Available on +3.3V Rail					**401.0**	mA	

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit	Source / note
Power Source Selection	9 V wall adapter (to J1)	MEAN WELL GST25A09-P1J	90-264 VAC	+9 V	2550	2550.0	mA	MEAN WELL GST25A datasheet: 9 V, 2.55 A, 23 W.
Loads Connected to This Source	+6 V rail through VR1 LM317T	ST LM317T	8.55 V in			925.3	mA	LM317 input current = output current + adjust current (100 uA).
	+5 V rail through VR2 L7805CV	ST L7805CV	8.55 V in			13.5	mA	Required +5 V current incl. margin.
	L7805CV quiescent current	ST L7805CV				8.0	mA	ST L78 datasheet: quiescent current 8 mA max.
Total Remaining Current Available on This Source						**1603.3**	mA	Subtracting the regulators' 1.5 A ratings instead (template method) would give 2550 - 3000 < 0; the real demand is far lower, so required currents are used.
Input fuse check	Total current through F1 (1 A slow-blow)					**946.8**	mA	Must stay below the 1000 mA fuse rating.

External Power Source 2	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit	Source / note
Power Source Selection	USB port of the PC (Curiosity Nano)	USB 2.0 port	5 V	+5 V	500	500.0	mA	USB 2.0 port: 500 mA. Nano VBUS PTC fuse 500 mA hold.
Loads Connected to This Source	+3.3 V rail through MIC5353	Microchip MIC5353	5 V in			99.0	mA	Required +3.3 V current incl. margin.
	nEDBG debugger + its regulator	PIC32MZ nEDBG	5 V			100.0	mA	Allowance for the on-board debugger (estimate; not given in the user guide).
Total Remaining Current Available on This Source						**301.0**	mA	

Thermal check

VR1 LM317T dissipation at worst case	**2.36**	W	(Vin - Vout) x I. Above ~1 W a TO-220 needs the heatsink (577102B00000G).
VR2 L7805CV dissipation	**0.08**	W	No heatsink needed.
FAN8100N output saturation loss at stall (approx.)	**0.62**	W	Vsat 0.55 V max at 0.4 A, scaled to 0.67 A; below the 2.0 W Pd rating.

Notes:

External supply voltage is set by the regulator dropout: LM317 needs about Vout + 2 V = 8 V; 9 V minus the 1N5819 drop (about 0.45 V) leaves 8.55 V.

The Curiosity Nano and the 9 V supply are separate sources; their grounds are tied on the board.

Motor stall is a short event (the firmware stops the motor at the end of travel); the normal running current is about 0.1-0.2 A.